

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K}$.

6.1. Determine the sum-of-angles formulas for sine and cosine using rotation matrices.

6.2. Determine the matrix form of the affine map obtained from a rotation with $\pi/3$ after a translation with $(-1, 0)$.

6.3. Verify that the matrices

$$A = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad C = \frac{1}{13} \begin{bmatrix} 5 & -12 \\ 12 & 5 \end{bmatrix} \quad \text{and} \quad D = \frac{1}{17} \begin{bmatrix} 8 & 15 \\ -15 & 8 \end{bmatrix}$$

belong to $\text{SO}(2)$. In each case determine the rotation angle. Moreover, check that when changing the sign of any row or column, the matrices are orthogonal but not special orthogonal.

6.4. For the matrices in the previous exercises, change the sign in the last column. Doing so you obtain orthogonal reflections. Determine the reflection lines.

6.5. Verify that the matrices

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \frac{1}{11} \begin{bmatrix} -9 & -2 & 6 \\ 6 & -6 & 7 \\ 2 & 9 & 6 \end{bmatrix}$$

belong to $\text{SO}(3)$.

6.6. Using rotation matrices deduce the parametrization $\gamma(t) = (t - \sin(t), 1 - \cos(t))$ obtained by rolling a circle of radius 1 on the x -axis.

6.1. Determine the sum-of-angles formulas for sine and cosine using rotation matrices.

A rotation with angle θ followed by a rotation with angle $\tilde{\theta}$ is the same as a rotation with angle $\theta + \tilde{\theta}$:

$$\text{Rot}_{\tilde{\theta}} \circ \text{Rot}_{\theta} = \text{Rot}_{\tilde{\theta} + \theta}$$

$$\begin{bmatrix} \cos \tilde{\theta} & -\sin \tilde{\theta} \\ \sin \tilde{\theta} & \cos \tilde{\theta} \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos(\tilde{\theta} + \theta) & -\sin(\tilde{\theta} + \theta) \\ \sin(\tilde{\theta} + \theta) & \cos(\tilde{\theta} + \theta) \end{bmatrix}$$

$$\begin{bmatrix} \cos \tilde{\theta} \cos \theta - \sin \tilde{\theta} \sin \theta & -(\cos \tilde{\theta} \sin \theta + \sin \tilde{\theta} \cos \theta) \\ \dots & \dots \end{bmatrix}$$

6.2. Determine the matrix form of the affine map obtained from a rotation with $\pi/3$ after a translation with $(-1, 0)$.

$$\begin{aligned} \phi(P) &= \text{Rot}_{\frac{\pi}{3}} \circ T_{(-1, 0)}(P) \\ &= \text{Rot}_{\frac{\pi}{3}} \left(P + \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \left(P + \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} P - \frac{1}{2} \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \end{aligned}$$

6.3. Verify that the matrices

$$A = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad C = \frac{1}{13} \begin{bmatrix} 5 & -12 \\ 12 & 5 \end{bmatrix} \quad \text{and} \quad D = \frac{1}{17} \begin{bmatrix} 8 & 15 \\ -15 & 8 \end{bmatrix}$$

belong to $SO(2)$. In each case determine the rotation angle. Moreover, check that when changing the sign of any row or column, the matrices are orthogonal but not special orthogonal.

A: Method I $A \cdot A^T = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix} = I_2$

$$\Rightarrow A \in O(2) \quad \Rightarrow A \in SO(2)$$

$$\det A = 1$$

Method II we see that $\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2 = 1$

$$\Rightarrow \exists \theta \text{ such that } \frac{4}{5} = \cos \theta \quad \frac{3}{5} = \sin \theta \text{ and } A = \text{Rot}_\theta$$

B: B is the matrix Rot_{90°

$$C: \left(\frac{5}{13}\right)^2 + \left(\frac{12}{13}\right)^2 = \frac{25 + 144}{169} = 1$$

$$D: \left(\frac{8}{17}\right)^2 + \left(\frac{15}{17}\right)^2 = \frac{64 + 225}{289} = 1$$

in all cases changing the sign of a column or row multiplies the determinant with -1 (properties of determinants)
but you can also check by direct calculations

6.4. For the matrices in the previous exercises, change the sign in the last column. Doing so you obtain orthogonal reflections. Determine the reflection lines.

A: $\tilde{A} = \frac{1}{5} \begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix}$

• For all cases we notice that the reflection line of the reflection $\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \tilde{A} \begin{bmatrix} x \\ y \end{bmatrix}$ is the set of points which don't move under the reflection, i.e. it is the set of solutions to

$$\tilde{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \Leftrightarrow \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Leftrightarrow x = 3y$$

$$\Rightarrow \text{the line is } x - 3y = 0 \quad \text{or} \quad \begin{cases} x = 3t \\ y = t \end{cases}$$

test: $\frac{1}{5} \begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \checkmark$

- In these cases the direction vectors for the line are eigenvectors for the eigenvalue 1.

B, C, D are similar just different calculations.

6.5. Verify that the matrices

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \frac{1}{11} \begin{bmatrix} -9 & -2 & 6 \\ 6 & -6 & 7 \\ 2 & 9 & 6 \end{bmatrix}$$

belong to $SO(3)$.

$A \in SO(3) \Leftrightarrow AA^T = I_3 \text{ \& } \det A = 1$

$$A^T = \frac{1}{3} \begin{bmatrix} -1 & -2 & -2 \\ 2 & -2 & 1 \\ -2 & -1 & 2 \end{bmatrix}$$

$$A \cdot A^T = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} -1 & -2 & -2 \\ 2 & -2 & 1 \\ -2 & -1 & 2 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

One checks that $\det A = 1$ so, yeah, $A \in SO(3)$

6.6. Using rotation matrices deduce the parametrization $\gamma(t) = (t - \sin(t), 1 - \cos(t))$ obtained by rolling a circle of radius 1 on the x -axis.

you can find this in the course material